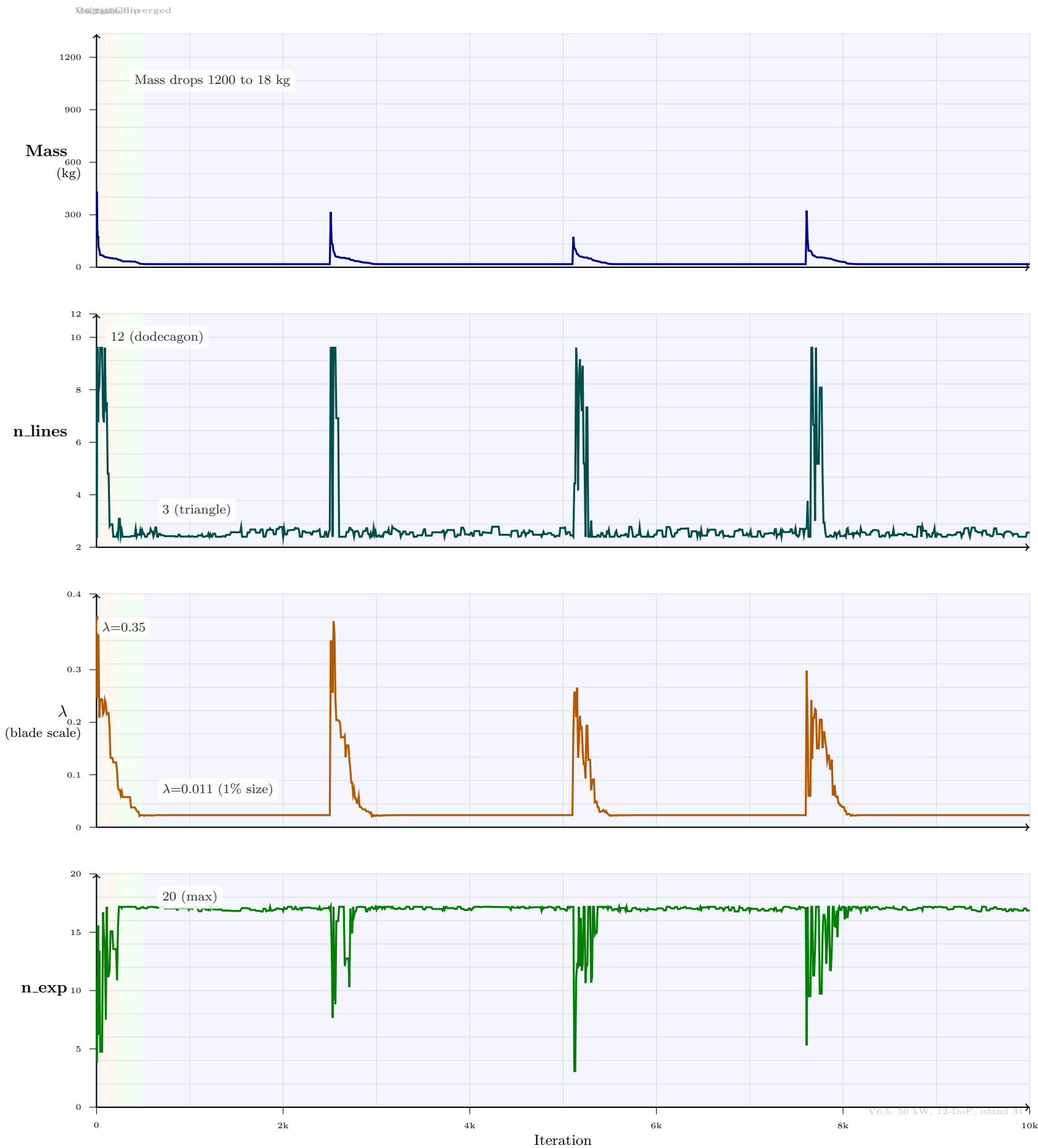


V6.5 Parameter Convergence – Island 31 (best)

50 kW, 12-DoF DE – 17.74 kg optimum



The cascade

1. Initial
n=3, thick beams (255 mm).
 $\lambda=0.35$, n_exp=4.
Big chunky design,
infeasible.

2. V6.2-like
Discovers n=12 dodecagon.
 $\lambda=0.22-0.28$, medium
blades (80–135 mm Do).
Feasible, but heavy.

3. Polygon flip
n drops 12→9→3.
 λ shrinks 0.21→0.01.
n_exp grows 8→20.
Thinner beams, more rotors.

4. Converged
n=3 triangle. $\lambda=0.011$.
n_exp=20 (at max bound).
Do=11 mm, r_hub=7.5 m.
17.74 kg, 57/60 islands.

17.74 kg

57/60 islands converged